Zadanie 1

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Zd.1

```
(.venv) debian@debian:~/Pulpit/UNIVERSITY$ pip3 install pandas
Requirement already satisfied: pandas in ./.venv/lib/python3.13/site-packages (2.3.3)
Requirement already satisfied: numpy>=1.26.0 in ./.venv/lib/python3.13/site-packages (from pandas) (2.3.5)
Requirement already satisfied: python-dateutil>=2.8.2 in ./.venv/lib/python3.13/site-packages (from pandas) (2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in ./.venv/lib/python3.13/site-packages (from pandas) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in ./.venv/lib/python3.13/site-packages (from pandas) (2025.2)
Requirement already satisfied: six>=1.5 in ./.venv/lib/python3.13/site-packages (from python-dateutil>=2.8.2->pandas) (1.17.0)

[notice] A new release of pip is available: 24.3.1 -> 26.0.1
[notice] To update, run: pip install --upgrade pip
(.venv) debian@debian:~/Pulpit/UNIVERSITY$
```

Zd.2

```
(.venv) debian@debian:~/Pulpit/UNIVERSITY$ python3
Python 3.13.5 (main, Jun 25 2025, 18:55:22) [GCC 14.2.0] on linux
Type "help", "copyright", "credits" or "license" for more information.
>>>
```

Zd.3

```
(.venv) debian@debian:~/Pulpit/UNIVERSITY$ python3
Python 3.13.5 (main, Jun 25 2025, 18:55:22) [GCC 14.2.0] on linux
Type "help", "copyright", "credits" or "license" for more information.
>>> lista = [1,2,3,4,5,6,7,8,9,10]
>>>
```

Zd.4

```
>>> lista2 = [1,2,3,4,5,"jeden","dwa","trzy",3.14,15.2]
```

Zd.5

```
>>> wyniki = []
```

```
>>> for elem in lista2:
...     try:
...         wyniki.append(int(elem))
...     except (ValueError, TypeError):
...         wyniki.append(False)
... print(wyniki)
...
[1, 2, 3, 4, 5, 1, 2, 3, 4, 5, False, False, False, 3, 15]
>>>
```

Zd.6

palindrom.py

```python
def is_palindrome():  1 usage
    word = input("Enter a word:")
    if word == word[::-1]:
        return True
    return False

print(is_palindrome())
```

```
/home/debian/Pulpit/UNIVERSITY/.v
Enter a word:ala
True
```

```
/home/debian/Pulpit/UNIVERSITY/.v
Enter a word:piwko
False
```

Zd.7

palindrom.py prime.py

```python
def is_prime(number):  2 usages
    if number < 2:
        return False
    for i in range(2, int(number ** 0.5) + 1):
        if number % i == 0:
            return False
    return True

print(is_prime(4))
print(is_prime(5))
```

```
/home/debian/Pulpit/UNIVERSITY
False
True
```

Zd.8

palindrom.py prime.py dataframe.py

```python
import pandas as pd

dane = {
    'Imię': ['Anna', 'Jan', 'Kasia', 'Piotr'],
    'Wiek': [25, 30, 28, 35],
    'Pensja': [5000, 6000, 4500, 7000],
    'Miasto': ['Warszawa', 'Kraków', 'Gdańsk', 'Wrocław']
}

df = pd.DataFrame(dane)

print("===== WYŚWIETLANIE =====")

print(df.head(2)) # wyświetla pierwsze n wierszy (bazowo 5)
print(df.shape) # wyświetla ile jest wierszy i kolumn
print(df.describe()) # wyświetla podstawowe statystyki
print(df["Imię"]) # wyświetla wartości kolumny "Imię"
print(df.loc[1,"Imię"]) #wyświetla pierwszą komórkę z kolumny "Imię"

print("\n\n\n===== AKTUALIZACJA WARTOŚCI =====")

df["Pensja"] = df["Pensja"] * 1.1  # +10% do pensji
print("Po podwyżkach:", df["Pensja"].tolist())

df.loc[df['Wiek'] < 30, 'Wiek'] = 18  # Wiek < 30 → 18
print("Po zmianie wieku:", df[['Imię', 'Wiek']])
```

```
===== WYŚWIETLANIE =====
   Imię  Wiek  Pensja    Miasto
0  Anna    25    5000  Warszawa
1   Jan    30    6000    Kraków
(4, 4)
            Wiek        Pensja
count   4.000000      4.000000
mean   29.500000   5625.000000
std     4.203173   1108.677891
min    25.000000   4500.000000
25%    27.250000   4875.000000
50%    29.000000   5500.000000
75%    31.250000   6250.000000
max    35.000000   7000.000000
0     Anna
1      Jan
2    Kasia
3    Piotr
Name: Imię, dtype: object
Jan



===== AKTUALIZACJA WARTOŚCI =====
Po podwyżkach: [5500.0, 6600.000000000001, 4950.0, 7700.000000000001]
Po zmianie wieku:      Imię  Wiek
0   Anna    18
1    Jan    30
2  Kasia    18
3  Piotr    35

Process finished with exit code 0
```